

Decent Coaching Classes

Std. VIII (CBSE) — Mind Your Math - I

Answer Key — Understanding Quadrilaterals (Parallelogram & Quadrilaterals)

Section A

- (1) (c) 360°
- (2) (b) equal
- (3) (a) equal
- (4) (b) bisect each other at right angles
- (5) (b) trapezium

Section B

- (1) Each exterior angle of a regular pentagon $= 360^\circ/5 = 72^\circ$
- (2) Sum of angles of a quadrilateral $= 360^\circ$. So, $100^\circ + 90^\circ + 70^\circ + x^\circ = 360^\circ \Rightarrow x = 100^\circ$
- (3) Adjacent angles of a parallelogram are supplementary. $2y + 3y = 180^\circ \Rightarrow y = 36^\circ$. Angles $= 72^\circ$ and 108°
- (4) Number of sides $= 360^\circ/45^\circ = 8$ (regular octagon)
- (5) $\angle C = \angle A = 70^\circ$ (opposite angles equal); $\angle B = \angle D = 180^\circ - 70^\circ = 110^\circ$ (adjacent angles supplementary)

Section C

- (1) Each interior angle of a regular hexagon $= (6-2) \times 180^\circ/6 = 720^\circ/6 = 120^\circ$
- (2) Perimeter $= 2(6 + 4.5) = 2 \times 10.5 = 21$ cm
- (3) $\angle A$ and $\angle B$ are adjacent angles, so they are supplementary: $3x + 2x + 10 = 180 \Rightarrow 5x = 170 \Rightarrow x = 34$. So $\angle A = 102^\circ$, $\angle B = 78^\circ$, and since opposite angles are equal, $\angle C = 102^\circ$, $\angle D = 78^\circ$.

Section D

- (1)(a) Let the angles be $3k, 4k, 5k, 6k$. Sum $= 18k = 360^\circ \Rightarrow k = 20^\circ$. Angles $= 60^\circ, 80^\circ, 100^\circ, 120^\circ$
- (1)(b) Diagonals of a parallelogram bisect each other, so $PR = 2 \times OP = 2 \times 3 = 6$ cm and $QS = 2 \times OQ = 2 \times 3.5 = 7$ cm